

ForgeIQ

Industrial Knowledge Intelligence Platform

Unified Asset & Operations Brain

An AI-powered operational intelligence platform built for the ET AI Hackathon 2026. ForgeIQ ingests and links heterogeneous industrial manuals, drawings, and logs, enabling domain RAG queries, real-time compliance checking, and predictive reliability analysis at the point of need.

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1. Executive Summary

ForgeIQ is an AI-powered Industrial Knowledge Intelligence platform designed to transform fragmented industrial documentation into a unified, queryable, and actionable "Operations Brain." By leveraging advanced Retrieval-Augmented Generation (RAG), Knowledge Graphs, and Multi-Agent AI systems, ForgeIQ connects engineering drawings, maintenance logs, operating procedures, and compliance standards. This integration reduces search times, mitigates the "knowledge cliff" from retiring workers, and minimizes unplanned downtime in heavy industries.

2. Problem Context & Business Impact

Asset-intensive industries suffer from severe information fragmentation:

- **Time Wastage:** Industry professionals spend up to 35% of their working hours searching for manuals, engineering diagrams, and maintenance history.
- **System Disconnection:** A typical plant operates 7 to 12 siloed document systems (P&IDs, compliance forms, incident reports, work orders).
- **Operational Costs:** This fragmentation contributes to 18-22% of unplanned downtime events due to decisions made without complete historical context.
- **Knowledge Cliff:** 25% of experienced industrial operators are retiring within the decade, taking undocumented expert knowledge with them.

ForgeIQ solves this by unifying all operational knowledge into a single point of intelligence.

3. System Architecture

ForgeIQ utilizes a split stack with modern AI frameworks:

- **Frontend:** Next.js 15, TypeScript, Tailwind CSS, Recharts. The layout is optimized for field technicians on mobile as well as plant engineers on desktops.
- **Backend:** FastAPI (Python 3.11), SQLAlchemy, PostgreSQL (relational metadata), ChromaDB (vector embeddings).
- **AI Layer:** Google Gemini via LangChain for context-aware answering and automatic relationship mapping.

- **Graph Engine:** NetworkX to model and visualize entities (Equipment, Locations, Personnel, Procedures) and their dependencies.
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4. Key Platform Modules

A. Universal Document Ingestion & Knowledge Graph

Ingests scanned forms, PDFs, CAD drawings, spreadsheets, and emails.

- Extracts critical tags: Equipment Tags (e.g. `P-101`), Process parameters (e.g. pressure, temperature), and regulatory references.
- Connects extracted entities into a semantic Knowledge Graph.

B. Expert Knowledge Copilot

RAG-powered conversational interface.

- Provides high-fidelity answers with specific page citations, text snippets, and relevance scores.
- Built to run smoothly on mobile devices for workers in the field.

C. Maintenance Intelligence & RCA Agent

Integrates work orders, inspection reports, and failure histories.

- Connects recurring failure patterns (e.g. centrifugal pump seal leaks).
- Auto-generates Root Cause Analysis (RCA) and predictive recommendations to lower downtime.

D. Quality & Regulatory Compliance Intelligence

Maps operational procedures against regulatory frameworks (API 510, OSHA 1910, NFPA 70E, ISO 45001).

- Automatically identifies compliance gaps.
- Generates audit-ready evidence packages.

E. Lessons Learned Engine

Scans incident history to flag hidden patterns.

- Calculates business impact (ROI, downtime cost, production loss).
- Pushes warnings to technicians before starting work on at-risk equipment.

5. Technical Verification & Performance

The platform builds and runs successfully:

- Frontend routes generate fully static pages via Next.js Turbopack.
- FastAPI backend starts up cleanly under Uvicorn.
- Full-stack containerization via Docker and Docker Compose ensures smooth deployment.